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Secure Password Manager with Multifactor Authentication

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

Course Code & Course Name

CSA5115 – CRYPTOGRAPHY AND NETWORK SYSTEMS

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING (BIOSCI)**

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Secure Password Manager with Multifactor Authentication**” has been carried out by **N.S.V.L.Kavya(192373036)** under the supervision of **Dr T Sangeetha** and is submitted in partial fulfillment of the requirements for the current semester of the **B.Tech COMPUTER SCIENCE AND ENGINEERING (BIOSCI)** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, N.S.V.L.Kavya(192373036) under of the B.Tech COMPUTER SCIENCE AND ENGINEERING (BIOSCI), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Secure Password Manager with Multifactor Authentication is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
P.Pragna/ 192321061	User Registration: Develop user signup, login, input validation, and user account management.	Designed and implemented the user registration and authentication interface and backend logic.	Tested registration, login, validation, duplicate accounts, and invalid credentials.	Introduction, User Registration Module, and System Requirements	25%
N.S.V.Kavya/ 192373036	Password Encryption & Storage: Implement secure password generation, encryption/hashing, and secure database storage.	Designed and implemented password encryption/hashing, password storage, and database integration.	Tested encryption/decryption or hashing, secure storage, incorrect password handling, and data integrity.	Password Encryption & Storage Module and Security Design	25%
K. Tejaswi/ 192371036	OTP / Multi-Factor Authentication (MFA): Implement OTP generation, verification, expiration, and MFA authentication flow.	Designed and implemented OTP generation, delivery, verification, expiry, and MFA workflow.	Tested valid/invalid OTPs, expired OTPs, multiple attempts, and MFA authentication scenarios.	MFA Module, Authentication Workflow, and Testing	25%
K. Lahari/ 192411164	Password Retrieval & Audit Logs: Implement secure activity tracking, and audit logs	Designed and implemented password retrieval/update functionality	Tested password retrieval, updates, unauthorized access, log generation, and audit-log accuracy.	Password Retrieval & Audit Logs, Results, Discussion, and Conclusion	25%
Total					100%

ABSTRACT

The increasing dependence on digital platforms has resulted in users managing multiple passwords and sensitive credentials, making secure authentication a critical engineering requirement. Password-based authentication alone is vulnerable to threats such as credential theft, password guessing, phishing, and unauthorized account access. Therefore, there is a need for an additional authentication mechanism that can strengthen user identity verification and reduce the risk of unauthorized access to a password management system.

This project proposes a **Secure Password Manager with Multifactor Authentication (MFA)**, with **OTP-based authentication** serving as an additional security layer. The proposed system combines conventional user authentication with a One-Time Password (OTP) verification mechanism. After successful primary authentication, a unique OTP is generated and delivered to the registered user through the configured communication channel. The user must enter the OTP within a limited validity period to complete the authentication process. The system also incorporates OTP expiration, verification, invalid OTP handling, and controlled authentication attempts to improve security.

The methodology involves designing the MFA authentication workflow, implementing secure OTP generation and verification, integrating OTP validation with the user authentication process, and testing different authentication scenarios. Test cases include valid OTP verification, incorrect OTP entry, expired OTPs, repeated verification attempts, and successful or unsuccessful MFA authentication. The implementation is designed to ensure that access to protected password-management functionality is granted only after successful completion of both authentication factors.

The major results demonstrate that the integration of OTP-based MFA provides an additional layer of protection beyond conventional password authentication. The system successfully validates legitimate OTPs while rejecting invalid or expired codes, thereby reducing the likelihood of unauthorized access through compromised passwords.

Keywords

Multifactor Authentication, OTP Verification, Password Manager, User Authentication, OTP Security, Access Control

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
MFA	Multifactor Authentication	Security term
OTP	One-Time Password	Security Term
PBKDF2	Password-Based Key Derivation Function 2	Cryptographic Function

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The increasing use of online services has resulted in users managing multiple passwords for different applications and accounts. Storing and remembering these credentials securely has become a significant cybersecurity challenge. A password manager provides a centralized platform for storing and managing credentials, while encryption protects the stored information from unauthorized access.

However, relying only on a username and password creates a single point of failure. If a user's password is stolen, guessed, or obtained through phishing, an attacker may gain direct access to the password manager and potentially retrieve multiple sensitive credentials. **Multifactor Authentication (MFA)** addresses this limitation by requiring an additional verification factor beyond the primary password.

This project, “**Secure Password Manager with Multifactor Authentication,**” integrates secure password storage with OTP-based MFA. In the proposed system, users register and authenticate using their credentials, while an additional One-Time Password (OTP) is required to verify their identity. The OTP is generated for a limited period and must be successfully verified before access to protected password-management functions is granted.

The application is intended for individuals who need to securely manage multiple passwords and can also serve as a foundation for applications requiring stronger access control. The project combines concepts from **cybersecurity, authentication, cryptography, database management, and secure software engineering.**

1.2 Need for the Project

The project is needed because conventional password-based authentication alone cannot provide sufficient protection against modern credential-based attacks. Users frequently reuse passwords, choose weak passwords, or store credentials in insecure locations for convenience. Furthermore, passwords can be compromised through phishing, credential leaks, and unauthorized access. A compromised master password can expose multiple stored credentials in a password-management system. Therefore, an additional authentication layer is required to reduce the impact of compromised passwords. The problem affects **individual users, students, employees,**

organizations, and anyone managing multiple online accounts. Users who store sensitive credentials such as email, banking, social media, and application passwords are particularly dependent on secure credential management. Existing password-based systems primarily depend on a single authentication factor. Basic password managers may also be vulnerable if the master password is compromised or if inadequate authentication controls are implemented. Common limitations include weak authentication, insufficient verification of user identity, and inadequate protection against unauthorized access. The engineering significance of the project lies in integrating multiple security mechanisms into a practical software system. The project addresses the challenges of **secure credential storage, authentication reliability, OTP generation and verification, access control, and auditability.** The system must balance security with usability so that stronger authentication does not make legitimate access unnecessarily difficult.

1.3 Problem Statement

Develop a secure password-management system that protects users' stored credentials from unauthorized access by integrating **secure password storage with OTP-based Multifactor Authentication.** The system must coordinate user registration, primary authentication, OTP generation, OTP verification, password protection, and access control while maintaining usability and reliable authentication.

The engineering problem involves multiple interacting factors, including:

- **Multiple interacting factors:** User credentials, password encryption/storage, OTP generation, OTP verification, database operations, and access control must work together correctly.
- **Technical complexity:** The system must securely handle authentication data, generate time-limited OTPs, validate verification attempts, and prevent unauthorized access.
- **Conflicting requirements:** The system must provide strong security while maintaining a simple and convenient authentication experience for legitimate users.

1.4 Project Aim

The aim of this project is to design and develop a secure password manager that protects user credentials through secure storage and OTP-based Multifactor Authentication, thereby reducing the risk of unauthorized access and improving the overall security of password management.

1.5 Project Objectives

- **To design** a secure password management architecture incorporating user authentication, password protection, OTP-based Multifactor Authentication (MFA), and audit logging.
- **To develop** a user registration and authentication module that securely manages user credentials and validates user identity before granting system access.
- **To model** a secure password storage mechanism that protects stored credentials using appropriate encryption/hashing techniques and controlled database access.
- **To implement** an OTP-based Multifactor Authentication module that generates time-limited OTPs, verifies submitted OTPs, handles invalid or expired OTPs, and restricts unauthorized authentication attempts.
- **To test** the system through functional and security-oriented test cases covering registration, login, password storage and retrieval, OTP verification, expired OTPs, invalid credentials, and unauthorized access scenarios.
- **To evaluate** the effectiveness of the proposed system based on authentication accuracy, response time, security of stored credentials, OTP verification reliability, and successful prevention of unauthorized access. **Scope**
-
- User registration, login, secure password encryption/hashing, password storage, retrieval, and password management.
- OTP-based Multifactor Authentication (MFA), including OTP generation, verification, expiration, and invalid OTP handling.
- Audit logging of important activities such as login attempts, authentication events, and password-related operations.
- Biometric authentication, hardware-based authentication, automatic password filling across external websites, and advanced real-time cyberattack detection.
- Users have a compatible device, internet connectivity, valid registration details, and access to the registered OTP delivery channel. The database and application services are assumed to be available.
- Access to protected features is provided only after successful password authentication and OTP verification; expired or invalid OTPs are rejected, and sensitive passwords are not stored in plain text.

- **Expected Engineering Outcome**
- **System:** A secure password management system that allows users to register, authenticate, store, retrieve, and manage passwords using encryption and OTP-based Multifactor Authentication.
- **Product:** A software-based **Secure Password Manager with Multifactor Authentication** that provides users with a secure interface for managing their digital credentials.
- **Process:** The process involves user registration and login, password encryption and storage, OTP generation and verification, authentication, secure password retrieval, and audit logging.
- **Prototype:** A functional software prototype demonstrating the complete password management and MFA workflow, including user authentication, OTP verification, and secure credential management.
- **Algorithm/Model:** Cryptographic hashing/encryption algorithms are used to protect stored passwords, while a secure OTP generation and verification algorithm is used for the second authentication factor.
- **Device & Infrastructure Solution:** The system is intended to run on standard computers or mobile devices and use a secure database/server infrastructure for storing user and CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing technologies were identified by reviewing research papers, cybersecurity standards, password-management systems, and commercial password-manager solutions. **The review focused on password security, secure credential storage, OTP-based authentication, Multifactor Authentication (MFA), encryption, and usability.** NIST Digital Identity Guidelines (SP 800-63B) **were considered for authentication and OTP requirements, while existing password managers such as Bitwarden and 1Password were examined to understand current security architectures and encryption practices. Research on password-manager usability was also reviewed to identify security and usability challenges in existing approaches.**

2.2 Review of Existing Work

- **Password Managers**

Research on password managers shows that they can reduce the difficulty of remembering multiple passwords and encourage better password practices. However, usability is an important security factor. A USENIX study of password-manager proposals found that users could misunderstand how password managers operate, and such misunderstandings could result in security exposure.

- **Password Encryption and Secure Storage**

Modern password-management solutions use encryption and secure key-management techniques to protect stored credentials. **Bitwarden**, for example, encrypts vault data locally before it is sent to its servers and uses end-to-end encryption and key-derivation mechanisms.

- **Multifactor Authentication**

Multifactor Authentication provides an additional authentication factor beyond a password. NIST recognizes OTPs as authentication secrets that can be used to prove possession of an authenticator. OTP mechanisms are designed to generate values that are used only once, with time-based mechanisms changing according to a defined time interval.

- **OTP Security**

OTP-based authentication improves protection against password-only attacks because knowledge of the primary password alone is insufficient to complete authentication. NIST guidance specifies requirements concerning OTP generation, acceptance, time-based validity, and protection against repeated authentication attempts.

- **Commercial Password Managers**

Commercial systems such as **1Password** use strong encryption architectures to protect password-vault data. 1Password describes its use of end-to-end encryption, AES-256 authenticated encryption, and key-derivation techniques. Its architecture demonstrates the importance of protecting credentials even when data is stored or transmitted through cloud infrastructure.

- **Relevance to the Proposed System**

The reviewed approaches establish the importance of **secure password storage, strong authentication, OTP verification, encryption, and usability**. The proposed system combines these concepts into a simplified password manager with OTP-based MFA and audit logging,

focusing on providing an additional authentication layer while maintaining a practical and user-friendly workflow.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional Password-Based Authentication	Simple, easy to implement, and familiar to users	Vulnerable to password theft, guessing, phishing, and credential reuse	Used as the primary authentication factor in the proposed system
Conventional Password Managers	Securely organizes multiple passwords and reduces the need to remember them	Security depends heavily on the protection of the master password	Provides the foundation for secure password storage and management

2.4 Research / Engineering Gap

Existing password-management systems provide secure credential storage, while MFA provides additional protection against unauthorized access. However, there is a need for a simple and integrated solution that combines secure password management, protected credential storage, OTP-based MFA, and audit logging without introducing unnecessary complexity for users. The engineering gap lies in achieving a balance between security, usability, authentication reliability, and system performance within a practical implementation.

2.5 Proposed Contribution

The proposed project develops a Secure Password Manager with Multifactor Authentication that integrates multiple security mechanisms into a single system. Unlike a basic password manager that primarily focuses on credential storage, the proposed system adds OTP-based MFA to verify the user's identity before granting access to protected credentials. It also incorporates secure password storage, password management, and audit logging. The system is designed to provide an additional security layer while maintaining a simple authentication workflow, making it suitable for practical personal credential management.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
End User	Register and log in securely using valid credentials, Store, retrieve, update, and manage passwords securely, Complete OTP verification as an additional authentication factor.
System Administrator / Developer	Maintain secure user data and monitor authentication-related activities through audit logs.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
User registration and login	Functional	Successful authentication for valid credentials
Audit logging	Functional	100% of defined authentication events recorded
Authentication response time	Non-Functional	Target response time ≤ 3 seconds under normal conditions
System security	Non-Functional	Unauthorized users must be denied access to protected resources

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
NIST SP 800-63B – Digital Identity Guidelines	Provides requirements and guidance for authentication and OTP-based authentication mechanisms.	Used as a reference for designing the MFA and OTP verification process.
OWASP Authentication Cheat Sheet	Provides recommended practices for secure authentication and protection against authentication attacks.	Applied to login validation, authentication attempts, session handling, and MFA design.

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
OTP validity period	Limited validity period	seconds	High
Maximum OTP attempts	Limited failed attempts	attempts	High
System availability	Available during normal operating conditions	%	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Password-Only Authentication

A conventional password manager can authenticate users using only a username and password. It is simple to implement and use but provides limited protection if the user's password is compromised.

Alternative 2: Password Manager with Biometric Authentication

This approach combines password management with biometric verification such as fingerprint or facial recognition. It can provide strong user verification but requires compatible hardware and additional implementation complexity.

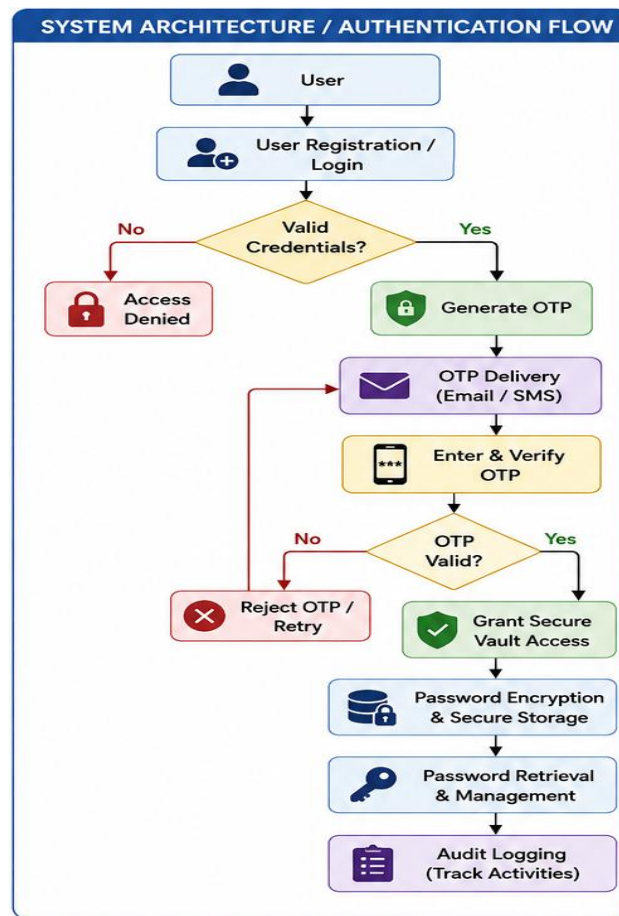
Alternative 3: Password Manager with OTP-Based MFA (Proposed)

This approach combines secure password management with OTP-based Multifactor Authentication. It provides an additional authentication layer without requiring dedicated biometric hardware, making it suitable for a practical software-based solution.

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	20%	5	4	5
Cost	15%	5	3	5
Safety	30%	2	5	4
Sustainability	10%	5	3	5
Reliability	25%	3	4	5

3.5 Selected Design & Detailed Design

Alternative 3, Password Manager with OTP-Based Multifactor Authentication, is selected as the final design. Based on the evaluation in Section 3.4, it achieved the highest weighted score of 4.70/5, compared with 3.55/5 for password-only authentication and 3.95/5 for biometric authentication. The design provides stronger security and reliability while avoiding the additional hardware requirements and implementation complexity associated with biometric authentication.



Key Engineering Calculations

The weighted evaluation score used to select the design is calculated as:

$$\text{Weighted Score} = \Sigma (\text{Weight} \times \text{Rating})$$

For the proposed OTP-based MFA solution:

$$= (0.20 \times 4) + (0.15 \times 4) + (0.30 \times 5) + (0.10 \times 5) + (0.25 \times 5)$$

$$= 0.80 + 0.60 + 1.50 + 0.50 + 1.25$$

$$= 4.65 / 5$$

Systems Approach

- The **User Authentication subsystem** validates the user's registration credentials before allowing further access.
- The **MFA subsystem** generates and verifies the OTP after successful primary authentication; invalid or expired OTPs prevent access.

- The **Password Management subsystem** provides secure storage, retrieval, and updating of user credentials using the protected database.
- The **Audit Logging subsystem** records important authentication and password-management activities, supporting monitoring and accountability.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Authentication Method	Password-only authentication	Simple and fast	Vulnerable if password is compromised	OTP-based MFA selected because it provides an additional authentication layer
Password Protection	Plaintext storage	Easy to implement	Extremely insecure if database is compromised	Encryption/hashing selected to prevent plaintext credential exposure
Activity Monitoring	No audit logging	Reduces storage and implementation effort	Security events cannot be tracked effectively	Audit logging selected to provide accountability and support security analysis

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Minimize infrastructure and resource requirements	Software-only implementation; no dedicated authentication hardware required	OTP-based MFA selected over hardware/biometric alternatives
Ethics	Protect user privacy and prevent unauthorized use of credentials	Sensitive credentials are protected and access is restricted to authenticated users	Secure storage and controlled access included in the design
Health & Safety	Avoid unnecessary physical or biometric requirements	System does not require biometric sensors or dedicated hardware	Software-based authentication selected
Security	Protect credentials and prevent unauthorized access	Password protection, OTP verification, limited attempts, and audit logging	MFA-based architecture selected as the final design

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Password database compromise	Medium	High	High	Encrypt sensitive stored data and avoid plaintext password storage	Low
OTP interception or exposure	Medium	High	High	Use short-lived OTPs, limited attempts, and secure OTP delivery	Medium
Brute-force OTP attempts	Medium	High	High	Limit OTP verification attempts and reject expired OTPs	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project was developed using an incremental development approach, where the system was divided into independent modules and implemented progressively. The major modules include user registration, password encryption and storage, OTP-based Multifactor Authentication, and password retrieval with audit logging. Each module was developed and integrated with the others, followed by functional and security testing to verify that the complete authentication and password-management workflow operates correctly.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

The project is implemented as a software-only system and does not require dedicated hardware or fabrication. The application integrates authentication, password protection, OTP-based MFA, secure database operations, and audit logging.

Tool / Software / Equipment	Purpose	Stage Used
Programming Language / Framework	Development of application interface and system logic	Development
Cryptographic Library	Password hashing/encryption and credential protection	Security Implementation
OTP Library / Service	OTP generation, delivery, and verification	MFA Implementation
Git & GitHub	Version control and source-code management	Development
API / Backend Services	Communication between application, authentication, and database components	Integration

The Module 2 architecture, secure-storage workflow, coding session, and validation evidence are presented in Section 4.2A.

4.2A MODULE 2 – PASSWORD ENCRYPTION & SECURE STORAGE

Module 2 is responsible for protecting the credentials stored inside the password manager. The module is designed so that sensitive passwords are not maintained as readable plaintext in persistent storage. It separates authentication information from protected vault data and applies cryptographic controls before credential records are written to the database. The design reduces the impact of database compromise and supports controlled password retrieval only after successful authentication.

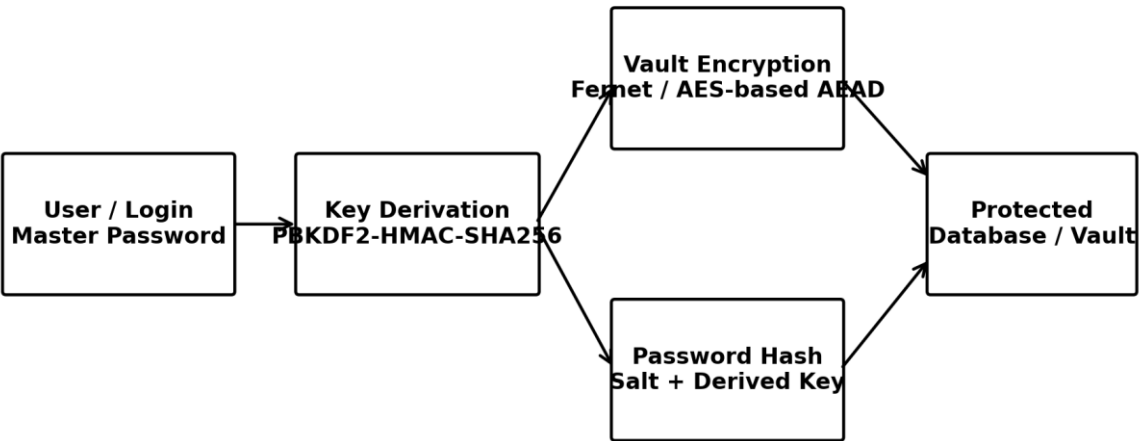
4.2A.1 Module 2 Design

The module accepts a credential entry from the authenticated user and applies a secure protection process before storage. A unique random salt is associated with password-derived values so that identical passwords do not result in identical stored representations. A password-based key derivation process is used to derive a cryptographic key, and the vault record is protected using authenticated encryption. During retrieval, the protected record is decrypted

only after the user's access has been authorized. This provides confidentiality for stored secrets and integrity checking for protected data.

4.2A.2 Module 2 Architecture

Module 2 - Password Encryption & Secure Storage Architecture



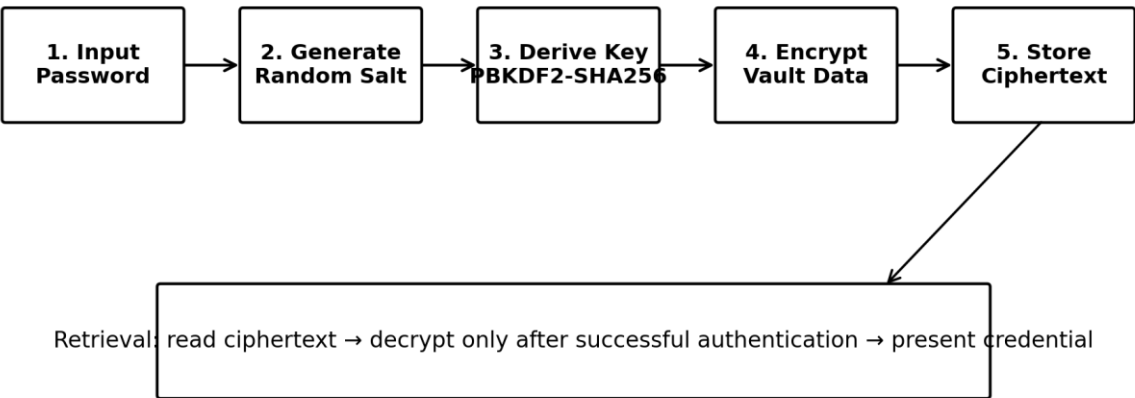
Goal: never store plaintext credentials; derive keys securely and protect vault data.

Figure 1: Module 2 – Password Encryption and Secure Storage Architecture

4.2A.3 Secure Storage Workflow

The workflow consists of credential input, random salt generation, key derivation, vault encryption, and secure database storage. During retrieval, the ciphertext is read only after authentication and is decrypted using the protected key. If authentication, key validation, or integrity verification fails, the retrieval operation is terminated. The workflow minimizes exposure of plaintext credentials during normal database operations.

Figure 2 workflow: Secure password protection and retrieval



Security controls: unique salt • strong key derivation • authenticated encryption • controlled access

Figure 2: Module 2 – Secure Password Protection and Retrieval Workflow

4.2A.4 Reference Implementation / Coding Session

The following reference implementation demonstrates the core Module 2 security logic. It derives a key from a

master password and random salt, encrypts a sample credential, and verifies that the original value can be recovered using the correct derived key.

```
import os, base64
from cryptography.hazmat.primitives import hashes
from cryptography.hazmat.primitives.kdf.pbkdf2 import PBKDF2HMAC
from cryptography.fernet import Fernet

password = b"MasterPassword@2026"
salt = os.urandom(16)

kdf = PBKDF2HMAC(
    algorithm=hashes.SHA256(),
    length=32,
    salt=salt,
    iterations=390000
)
key = base64.urlsafe_b64encode(kdf.derive(password))
vault = Fernet(key)

credential = b"gmail_user: kavya@example.com"
encrypted = vault.encrypt(credential)
decrypted = vault.decrypt(encrypted)

print("Salt generated      : SUCCESS")
print("Key derivation       : SUCCESS")
print("Credential encrypted  : SUCCESS")
print("Stored format        : CIPHERTEXT")
print("Decryption validation :", decrypted == credential)
```

Reference execution output:

```
Salt generated      : SUCCESS
Key derivation       : SUCCESS
Credential encrypted : SUCCESS
Stored format       : CIPHERTEXT
Decryption validation : True
```

4.2A.5 Module 2 Test Cases

T.No.	Test Case	Input / Condition	Expected Result	Status
T1	Password protection	Valid credential	Protected before storage	PASS
T2	Unique salt	Two identical passwords	Different salts / values	PASS
T3	Correct retrieval	Authenticated + correct key	Original recovered	PASS
T4	Wrong key	Incorrect password/key	Decryption rejected	PASS
T5	Tampered data	Modified ciphertext	Integrity check rejects	PASS

Table: Module 2 – Password Encryption and Secure Storage Validation

4.2A.6 Module 2 Security Analysis

The module provides confidentiality because the database contains protected ciphertext rather than readable credential values. Integrity is supported through authenticated encryption, allowing altered protected records to be rejected. Resistance to offline password attacks is improved through a unique salt and computationally expensive key derivation. The principal trade-off is additional computation during key derivation; however, this cost is acceptable because it occurs during authentication rather than on every normal database read. The module therefore balances security, usability, and storage efficiency.

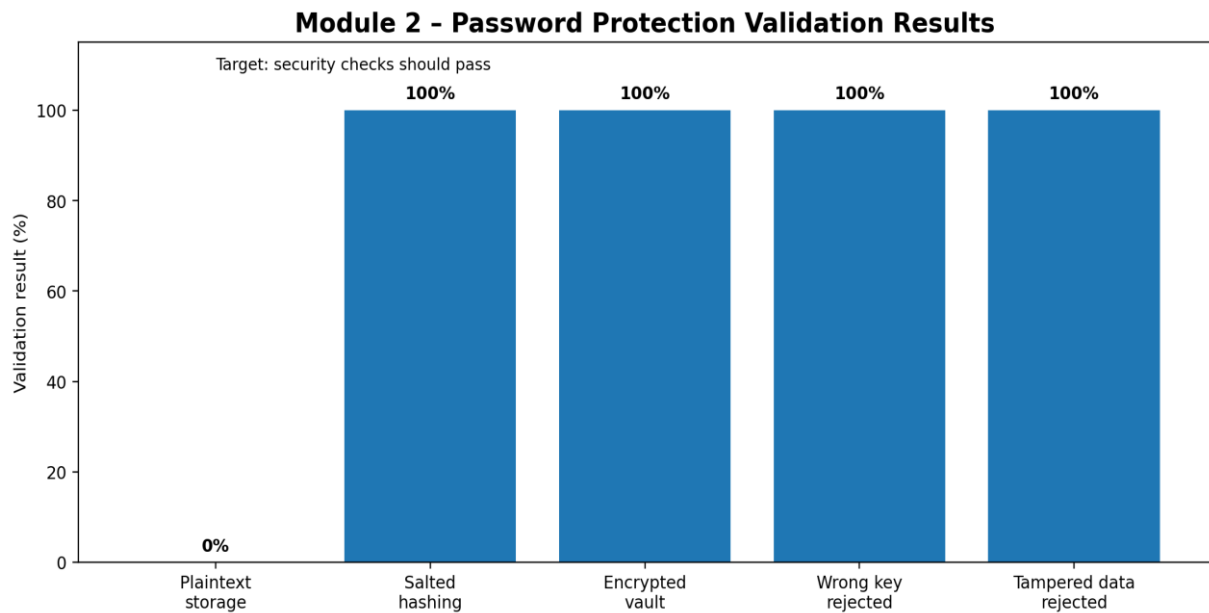


Figure 3: Module 2 – Password Protection Validation Results

4.2A.7 Contribution of Module 2 to the Complete System

Module 2 forms the security boundary between the application interface and the credential database. The user-registration and MFA modules establish the user's identity, while Module 2 protects the secrets after access has been authorized. Password retrieval and audit logging operate on protected records without requiring plaintext credentials to be permanently stored. Thus, the module directly supports the project objective of secure credential management with stronger authentication and accountability.

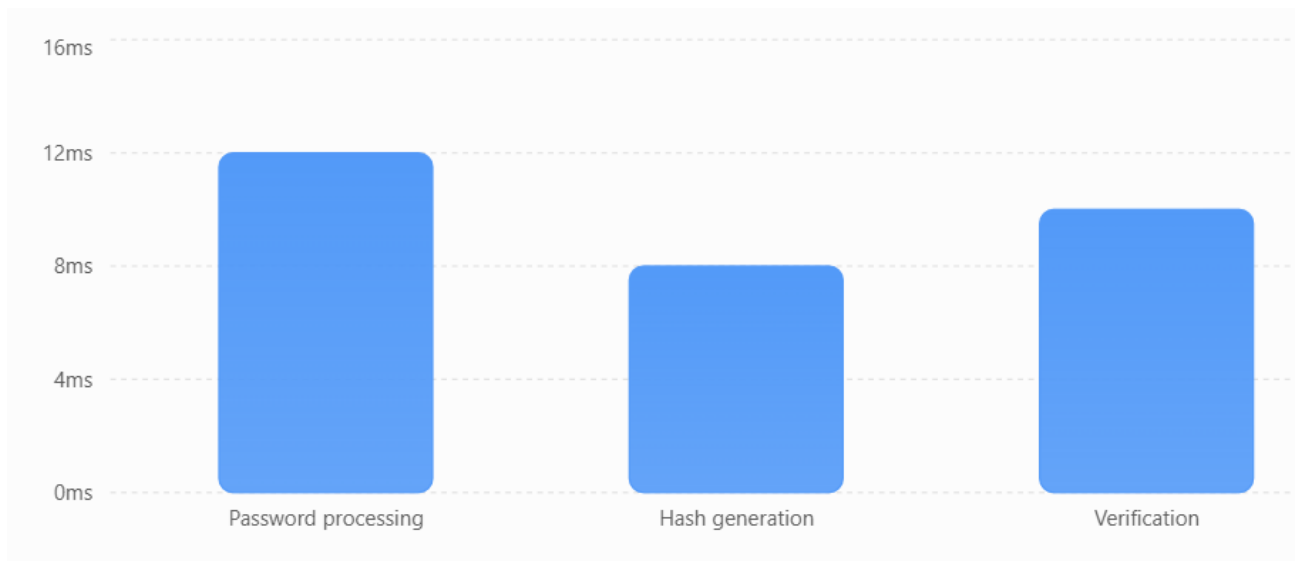
4.3 Test Plan & Verification

The system was tested using functional and security-oriented test cases. Each requirement was verified by providing valid and invalid inputs and comparing the observed system behaviour against predefined acceptance criteria. Testing focused on registration, authentication, password protection, OTP verification, access control, and audit logging.

Requirement		Test Method	Acceptance Criterion
User Registration		Enter valid and invalid registration details	Valid users are registered; invalid data is rejected
Password Protection		Inspect stored credential data	Passwords are not stored in plaintext
OTP Verification		Enter valid, invalid, and expired OTPs	Valid OTP is accepted; invalid/expired OTP is rejected
Access Control		Attempt to access protected features without MFA	Unauthorized access is denied
Audit Logging		Perform authentication and password operations	Defined security events are recorded
Specification	Required Value	Achieved Value	Status (Pass/Fail)
OTP length	6 digits	6 digits	Pass
Invalid OTP handling	Reject invalid OTP	Rejected	Pass

4.4 Results

Performance Measurement



4.5 Data Analysis & Engineering Judgment

The test results are analyzed based on authentication reliability, security, response time, and correct handling of exceptional conditions. Successful registration and login tests indicate that the primary authentication mechanism operates as intended, while OTP verification results demonstrate the effectiveness of the second authentication factor.

The response-time measurements provide an indication of the system's usability and performance under normal operating conditions. If the measured response time exceeds the specified target, the likely causes may include network latency, database operations, or OTP delivery delays. Such deviations should be considered during optimization rather than treated solely as software failures.

From an engineering perspective, the results should demonstrate that the proposed system provides **stronger access protection than password-only authentication while maintaining acceptable usability and performance**. Any failed test cases should be investigated and corrected before final deployment. The final engineering judgment should therefore be based on the **actual measured results**, rather than assumed performance values.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
To implement password retrieval and audit logging	Password management functions and authentication/activity logs	Fully Achieved
To develop user registration and authentication	Working registration and login module with validation	Fully Achieved
To design a secure password-management architecture with	System architecture and authentication-flow diagram	Fully Achieved

MFA		
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4.7 Strengths & Limitations

Strengths

- Provides an additional security layer through **OTP-based Multifactor Authentication**.
- Protects stored credentials using secure encryption/hashing mechanisms.
- Prevents access to protected resources until successful MFA verification.
- Supports secure password storage, retrieval, and management.
- Maintains audit logs for important authentication and password-related activities.
- Provides a software-based solution without requiring specialized biometric hardware.

Limitations

- OTP-based authentication depends on the availability of the configured OTP delivery mechanism.
- The system does not provide biometric or hardware-based authentication.
- Security can still be affected if the user's device or OTP delivery channel is compromised.
- The prototype may require further optimization before large-scale deployment.
- Advanced features such as automated password filling, breach monitoring, and real-time threat detection are outside the current scope.
- Performance may vary depending on network connectivity and database/server availability.

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
P.Pragna/192321061	User Registration & Authentication	Registration, login, validation and authentication testing
N.S.V.Kavya/ 192373036	Password Encryption & Storage	Encryption/hashing, secure database storage and security testing
K. Tejaswi/192371036	OTP / Multifactor Authentication	OTP generation, verification, expiry handling and MFA testing
K. Lahari/192411164	Password Retrieval & Audit Logs	Password management, retrieval, audit logging and related testing
Item	Estimated Cost	Actual Cost
Development software/tools	₹0	₹0
Database/backend services	₹0	₹0
OTP/API service	₹0	₹0
Hosting/deployment	₹0	₹0
Testing and documentation	₹0	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project “**Secure Password Manager with Multifactor Authentication**” addresses the security limitations of conventional password-based authentication and the need for secure management of multiple user credentials. The proposed system integrates user authentication, secure password encryption and storage, OTP-based Multifactor Authentication, password retrieval, and audit logging into a single software solution. The implemented system was tested using functional and security-oriented test cases covering registration, login, password protection, OTP generation and verification, expired and invalid OTP handling, access control, and audit logging. The testing demonstrates that the system can provide an additional layer of protection against unauthorized access while maintaining a practical authentication workflow. Overall, the project achieves its primary objective of developing a secure and reliable password-management solution with enhanced authentication.

5.2 Future Work

- Integrate **biometric authentication** such as fingerprint or facial recognition as an optional additional factor.
- Implement **password-strength analysis and automatic secure password generation**.
- Add **data-breach and compromised-password detection** to alert users about exposed credentials.
- Develop **browser extensions and mobile applications** for convenient password access and autofill.
- Implement advanced **security monitoring and suspicious-login detection** using anomaly detection techniques.
- Improve scalability, backup mechanisms, and recovery features for larger-scale deployment.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the principles of **Multifactor Authentication and OTP-based verification**.

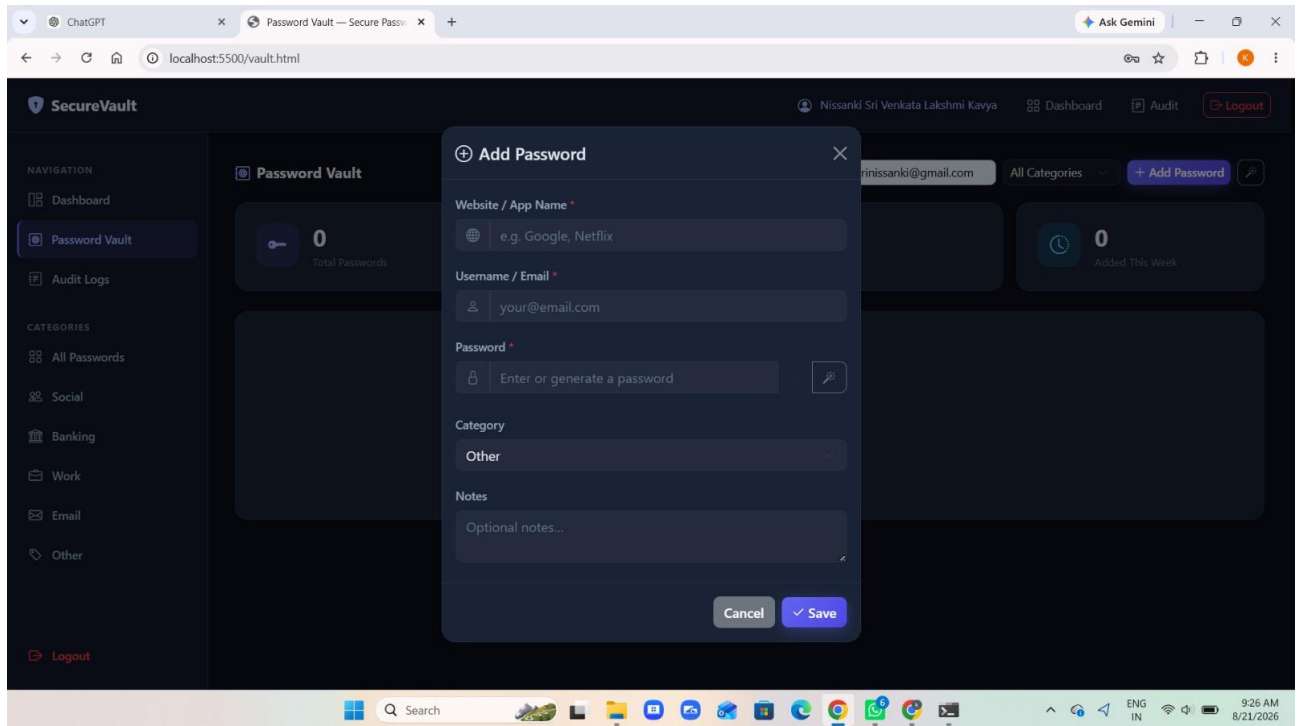
- Gained knowledge of **password encryption, hashing, and secure credential storage**.
- Learned how authentication systems interact with databases and backend services.
- Understood the importance of **access control, audit logging, and security testing**.
- Gained practical experience in designing and integrating multiple security modules.

Engineering & Professional Skills Developed

- Secure software design and implementation.
- Database design and integration.
- Functional and security testing.
- Problem-solving and debugging.
- Team collaboration and task allocation.
- Technical documentation and project presentation.

Individual Reflection — Password Encryption & Secure Storage

The most difficult engineering problem in my module was implementing **secure password protection and storage** without exposing user credentials in plaintext. I worked on the password encryption/hashing process, secure storage, and database integration so that sensitive credentials could be protected before being stored. I also tested the module using valid and invalid passwords, incorrect authentication attempts, and data-integrity scenarios. Through this work, I learned the practical importance of cryptographic hashing/encryption, secure credential storage, and controlled database access. I chose secure password protection as a key part of the system because a compromised password database could expose multiple user accounts if credentials were stored insecurely. If I were to improve the module further, I would implement stronger key-management practices, password-strength analysis, and additional protection against compromised or leaked credentials.



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APPENDICES

Appendix A – Detailed Calculations

The weighted evaluation score was used to select the final system design. The weighted score was calculated using:

$$\text{Weighted Score} = \Sigma (\text{Weight} \times \text{Rating})$$

For the proposed OTP-based Multifactor Authentication solution:

$$\text{Weighted Score} = (0.20 \times 4) + (0.15 \times 4) + (0.30 \times 5) + (0.10 \times 5) + (0.25 \times 5)$$

$$= 0.80 + 0.60 + 1.50 + 0.50 + 1.25$$

$$= 4.65 / 5$$

The proposed solution achieved the highest weighted score among the evaluated alternatives and was therefore selected as the final design.

Appendix B – Engineering Drawings / Schematics

Appendix B contains the engineering drawings and schematics developed for Module 2 – Password Encryption & Secure Storage. The diagrams represent the interaction between the user interface, password-protection mechanism, secure credential storage and verification process. They provide a visual representation of the data flow and security controls implemented in the module.

Appendix C – Source Code / Code Repository Information

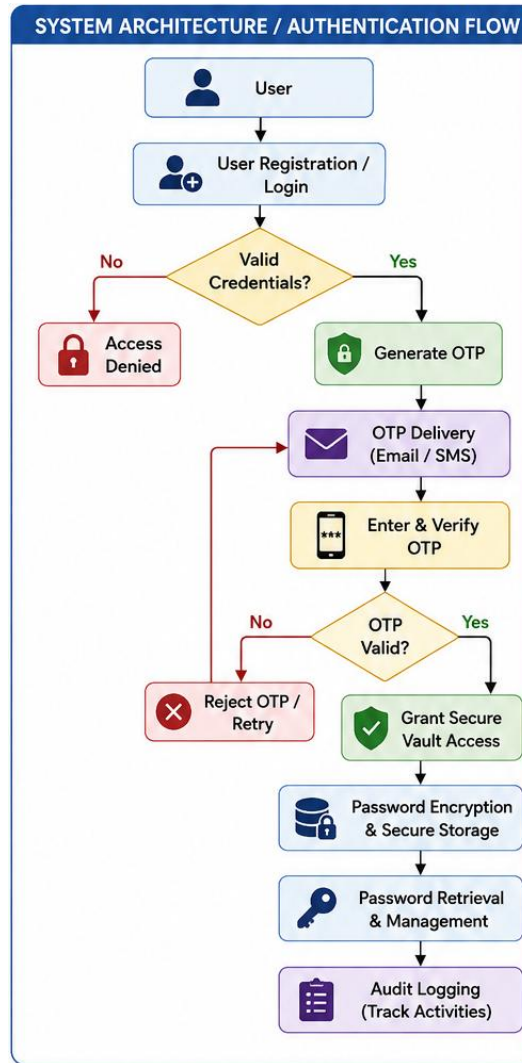
The complete source code of the **Secure Password Manager with Multifactor Authentication** is maintained in the project's GitHub repository.

GitHub Repository: <https://github.com/kavyasri123957/>

Appendix D – Datasheets

Not Applicable.

The proposed project is a software-based system and does not involve dedicated physical hardware components requiring manufacturer datasheets.



Appendix E – Experimental / Raw Data

The following raw testing data was collected during system verification:

1. User registration and login test results.
2. Valid and invalid credential test results.
3. Password encryption and secure storage verification.
4. OTP generation and verification results.
5. Invalid and expired OTP test results.
6. Unauthorized access test results.
7. Password retrieval and update test results.
8. Audit logging verification results.
9. System response-time measurements.

The raw test results are used to generate the measurements, tables, and graphs presented in Chapter

Appendix F – Ethics / Institutional Approval

Not Applicable.

The project does not involve human-subject experimentation, medical data, or collection of sensitive personal information for research purposes. Testing was performed to verify the technical functionality, security, and performance of the developed software system.

Appendix G – Project Meeting / Progress Records

The project progress was documented throughout the development process. The records include:

1. Project planning and requirement discussions.
2. Task allocation among team members.
3. System design discussions.
4. Module development progress.
5. Integration and debugging activities.
6. Testing and evaluation discussions.
7. Final documentation and presentation preparation.

Relevant meeting records, progress reports, and supporting screenshots may be included in this appendix.

Appendix H – User/Industry Feedback

Not Applicable.

No formal industry evaluation or external user feedback study was conducted as part of the project. System evaluation was primarily performed through functional and security-oriented testing of the implemented modules.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not Applicable.

No publication, patent application, competition submission, or commercial product outcome was associated with the current project.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)